AFFINE SCHEMES HAVE FINITE LIMITS COMPUTED BY TENSOR, MATCHING LOCALLY RINGED SPACES

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Editorial note. Counted span is the author's own proof environment, source lines 1220-1243, following the statement at 1201-1218.

Editorial note. The Mor-set chain is displayed line by line; each equality is an application of the named adjunction.

Allowed standard prerequisites (editorial).

- Morphisms into an affine scheme. lemma-morphism-into-affine (same chapter, cited at ~1221). Gives Mor(X, Spec(A)) = Hom(A, O_X(X)) and Spec(Z) final.

Source reference. Original reference 001U: focused-seeds/geometry/S9-schemes-fibre-product-affine-schemes/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 01I1: sources/stacks-schemes/pinned-original-source.tex, source lines 1058–1058. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *The category of affine schemes has finite products, and fibre products. In other words, it has finite limits. Moreover, the products and fibre products in the category of affine schemes are the same as in the category of locally ringed spaces. In a formula, we have (in the category of locally ringed spaces)*

$$\mathrm{Spec}(R) \times \mathrm{Spec}(S) = \mathrm{Spec}(R \otimes_{\mathbf{Z}} S)$$

and given ring maps $R \to A$, $R \to B$ we have

$$\mathrm{Spec}(A) \times_{\mathrm{Spec}(R)} \mathrm{Spec}(B) = \mathrm{Spec}(A \otimes_R B).$$

Proof. This is just an application of Lemma 01I1. First of all, by that lemma, the affine scheme $\mathrm{Spec}(\mathbf{Z})$ is the final object in the category of locally ringed spaces. Thus the first displayed formula follows from the second. To prove the second note that for any locally ringed space X we have

$$\begin{aligned} \mathrm{Mor}(X, \mathrm{Spec}(A \otimes_R B)) &= \mathrm{Hom}(A \otimes_R B, \mathcal{O}_X(X)) \\ &= \mathrm{Hom}(A, \mathcal{O}_X(X)) \times_{\mathrm{Hom}(R, \mathcal{O}_X(X))} \mathrm{Hom}(B, \mathcal{O}_X(X)) \\ &= \mathrm{Mor}(X, \mathrm{Spec}(A)) \times_{\mathrm{Mor}(X, \mathrm{Spec}(R))} \mathrm{Mor}(X, \mathrm{Spec}(B)) \end{aligned}$$

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which proves the formula. See Categories, Section 001U for the relevant definitions. □